

LUIS ANGULO

Tucson, AZ | 520-602-7448 | luisferangulob@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

University of Arizona, College of Science

Bachelor of Science in Data Science — Comprehensive Statistics Emphasis
Minors in Computer Science, Microbiology

Tucson, Arizona

Expected May 2027

GPA: 3.85

Research Experience

Computational Plant Sciences Laboratory

Quantitative Modeling Research Assistant

Tucson, Arizona

December 2025 – Present

- Engineered Python simulation pipelines for a 4.9-million-run Monte Carlo campaign of stochastic 3D root architectures, automating parameter sweeps, geometry generation, checkpointing, and high-volume output collection.
- Developed reproducible Python and R workflows computing 1,073 quantitative scientific outputs across morphology, geometry, branching, resource capture, generation, and Horton-Strahler structure.
- Built a Streamlit/Plotly research application integrating principal component analysis (PCA) to compare simulated root architectures, identify structurally similar systems, and analyze 5 reproducible model replicates per configuration.

Lunar and Planetary Laboratory,

Undergraduate Researcher

Tucson, Arizona

August 2026 – Present

- Developing statistical and machine-learning workflows to analyze Martian crater diameter and spatial-distribution data associated with debris-covered glaciers and other glacial landforms.
- Applying clustering and exploratory data-analysis methods to characterize crater populations and support relative age estimation of Martian glaciers from crater abundance and surface characteristics.

AI in Healthcare Simulation Education and Training VIP,

Vertically Integrated Project Researcher

Tucson, Arizona

May 2025 – August 2025

- Contributed to development of interactive AI-driven virtual patients with distinct diseases, symptoms, and evolving clinical states for medical-student diagnostic training.
- Helped implement multi-stage diagnostic scenarios in which learner decisions altered patient progression, including worsening conditions following incorrect diagnoses or treatment choices.
- Integrated application components through APIs and contributed programming in Python and C++ to support patient behavior, simulation logic, and data exchange.

Department of Chemistry & Biochemistry, University of Arizona

Undergraduate Researcher

Tucson, Arizona

January 2024 – May 2024

- Conducted multiscale computational analysis of 3 OLED molecules—3phCN-ICz, CzPA, and DOBNA—using quantum calculations and molecular dynamics to evaluate electronic properties, molecular interactions, and optical-gap behavior.
- Performed 8,000-ps molecular dynamics simulations under NVT/NPT conditions using GROMACS and OPLS-AA, alongside Gaussian B3LYP/6-31G(d,p) calculations of HOMO, LUMO, and fundamental-gap values.
- Analyzed 5 time-resolved molecular frames per compound using Python, pandas, NumPy, and matplotlib to compare energy, temperature, and electronic-property changes against published experimental results.

Professional Experience

Office of Research & Partnerships

Data Analyst

Tucson, Arizona

May 2024 – Present

- Analyze and organize research-development data from internal systems and APIs to support KPI development, departmental reporting, faculty-engagement metrics, and interactive dashboards.
- Develop Python-based automation tools, data-processing workflows, and internal applications to streamline recurring office tasks and improve consistency across reporting processes.
- Maintain and expand departmental web resources using HTML/CSS while contributing to data-collection infrastructure and centralized database planning for longitudinal office metrics.

Oxford Global Resources

Technology Consulting Intern

Phoenix, Arizona

May 2026 – August 2026

- Used SQL and AWS-based data environments to support database organization, querying, and optimization for client product, sales, and operational data.
- Built and analyzed Power BI dashboards to translate client data into business metrics, trends, and reporting insights.
- Supported technology-consulting projects involving database design, web-process optimization, and digital-system improvements in collaboration with technical and marketing teams.

Leadership Experience

Quantitative Finance Association, Eller College of Management

Vice President of Technology

Tucson, Arizona

January 2026 – Present

- Lead technical workshops on Python-based quantitative trading, covering market-data APIs, financial feature engineering, machine learning, and algorithmic strategy development.
- Developed instructional trading models using yfinance, pandas, NumPy, scikit-learn, and TensorFlow/Keras LSTMs to model short-term stock returns.
- Demonstrated end-to-end trading pipelines using the Alpaca API, including backtesting, portfolio rebalancing, SPY benchmark comparison, and strategy-correlation analysis.

HackArizona

Director of Internal & External Outreach

Tucson, Arizona

May 2026 – Present

- Coordinate outreach to companies, university units, local businesses, and community partners to secure sponsorships, in-kind contributions, judges, and technology partners for HackArizona programming.
- Manage partner communication and participation logistics across sponsorship, judging, event support, and technical collaboration.
- Develop and maintain outreach pipelines for prospective partners while coordinating with the organizing team on external engagement strategy.

Technical Skills

Programming: Python, R, SQL, Java, C++, JavaScript, HTML/CSS, MATLAB

Data Science & Machine Learning: pandas, NumPy, scikit-learn, TensorFlow/Keras, Power BI, Tableau

Scientific Computing: Linux, Slurm/HPC, GROMACS, Gaussian/GaussView, Streamlit, Plotly, Git/GitHub

Cloud & Development: AWS, APIs, PySide6/Qt, OpenCV

Honors and Awards

Global Wildcat Tuition Award, University of Arizona — \$18,500

Dean's List, University of Arizona — Spring 2024

Dean's List, University of Arizona — Fall 2024